Learning Spectral Initializations for Low-Rank Attention

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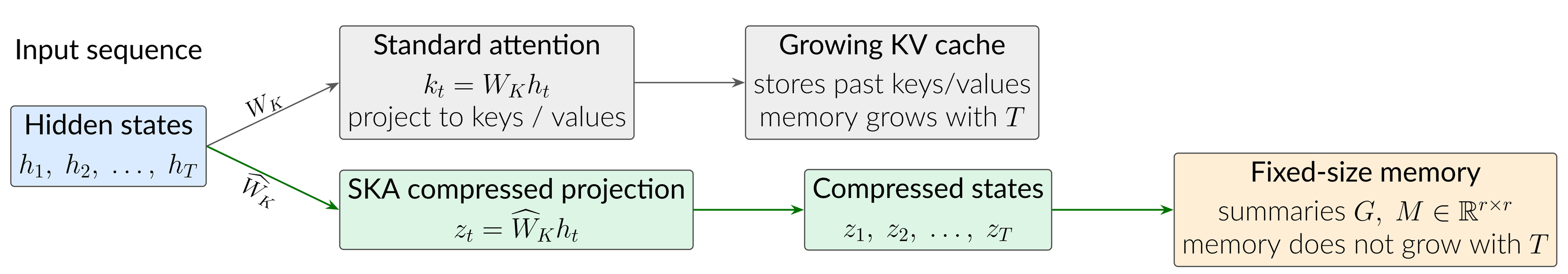
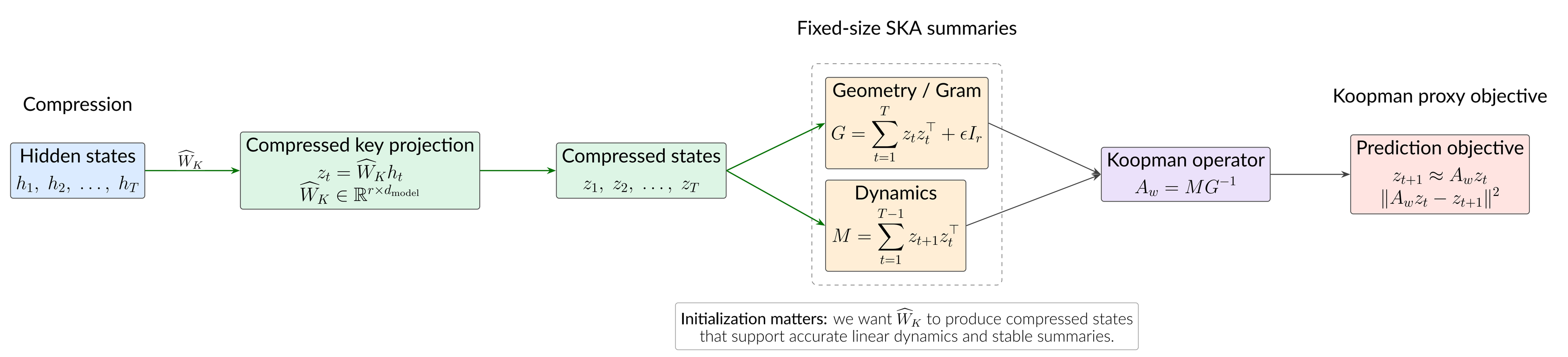
Motivation

- **Problem**: Standard attention requires a KV cache that grows with sequence length.
- **Fix**: SKA [4] replaces attention layers with fixed-size memory modules for constant-memory inference without sacrificing attention-like recall.
- **Key object**: compressed key projection $\hat{W}_K$.
- **Goal**: compare fixed and learned initialization strategies for $\hat{W}_K$.
- **Result**: learned strategies improve the proxy objective; downstream gains are rank-dependent.

Background: Spectral Koopman Attention





- **Why initialization matters**: $\hat{W}_K$ controls both Koopman prediction loss and numerical conditioning.

Background: Eckart-Young Theorem

- Current SKA uses **random orthogonal** initialization: perfectly-conditioned, but not aligned with pretrained W_K.
- **Eckart–Young [2]**: for any rank-r matrix B,

$$U_r \Sigma_r V_r^T = \arg \min_{\text{rank}(B) \le r} \|W_K - B\|_F.$$

- Thus, truncated SVD preserves as much of W_K as possible among rank-r reconstructions.
- **Question**: can learned singular-value scaling improve over fixed SVD baselines?

Datasets

- **Data/model**: WikiText-103 [3] with pretrained Pythia-160M [1]; sequences tokenized to length 256.
- **Koopman proxy**: extract Layer 2 hidden states h_t and evaluate compressed states $z_t = \hat{W}_K h_t$ on 512 train, 256 validation, and 256 test sequences (learning parameters/fitting A_w, cross validation, evaluation, respectively)
- **Downstream task**: replace Layer 2 attention with SKA and evaluate next-token LM loss and perplexity.

Methods

Strategy	Singular-value scaling rule
SVD-noscale	$g(\sigma_i) = 1$
SVD-sqrt	$g(\sigma_i) = \sigma_i^{1/2}$
SVD-full	$g(\sigma_i) = \sigma_i$
learned_b	$g_b(\sigma_i) = \sigma_i^b$
learned_a,b	$g_{a,b}(\sigma_i) = a\sigma_i^b$
MLP spectral scaling	$g_i = \sigma_i^{0.5} \exp(\alpha f_\theta(\log \sigma_i, i/r))$

Table 1. Initialization strategies for constructing $\widehat{W}_K = g(\Sigma_r) V_r^\top$.

- Scaling objective

$$\mathcal{L}(\theta) = \frac{1}{T-1} \sum_{t=1}^{T-1} \|A_w z_t - z_{t+1}\|_2^2 + \lambda \log \kappa(G), \qquad \kappa(G) = \frac{\sigma_{\max}(G)}{\sigma_{\min}(G)}.$$

- Evaluation metrics.

Metric	Formula	Interpretation
Relative MSE	$\frac{\sum_{t=1}^{T-1} \|A_w z_t - z_{t+1}\|_2^2}{\sum_{t=1}^{T-1} \|z_{t+1}\|_2^2 + \epsilon}$	Koopman prediction error; lower is better
Condition number	$\kappa(G) = \frac{\sigma_{\max}(G)}{\sigma_{\min}(G)}$	Numerical stability; lower is better
LM loss ($\mathcal{L}_{LM}$)	$-\frac{1}{T-1} \sum_{t=1}^{T-1} \log p(w_{t+1} \mid w_{\le t})$	Next-token prediction error; lower is better
Perplexity	$\text{PPL} = \exp(\mathcal{L}_{LM})$	Exponentiated LM loss; lower is better

Experiments/Results - Koopman Proxy

- **Goal**: Learn singular-value scaling parameters (via optimizing $\mathcal{L}(\theta)$) and fit A_w on the training set for each strategy.
- **Hyperparameters**: Select λ and α using 5-fold cross validation.
- **Evaluation**: Compare relative MSE and conditioning across ranks $8, 16, 32, 64$.

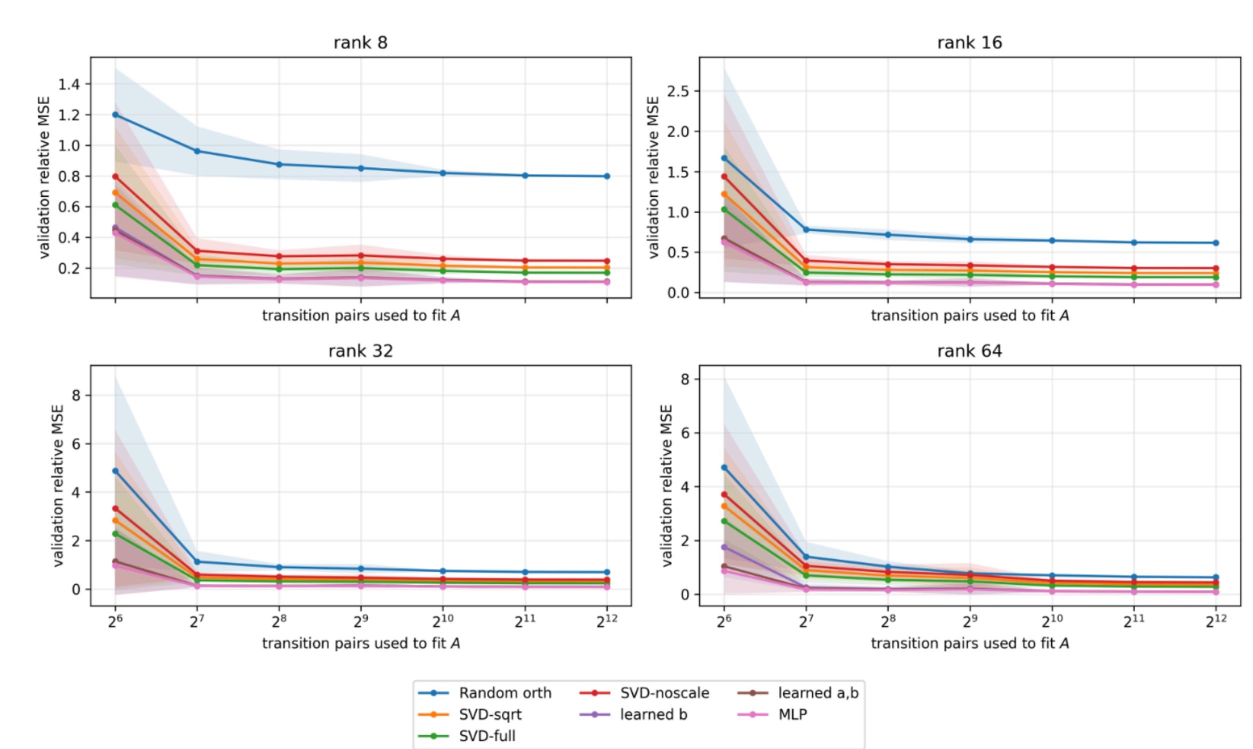


Figure 1. Koopman proxy MSE vs fitted transition pairs.

$\kappa(G)$ per strategy	r=8	r=16	r=32	r=64
Random orth.	1.3	1.8	2.1	2.6
SVD-noscale	2.1	2.2	2.5	2.7
SVD-sqrt	2.3	2.4	2.7	2.9
SVD-full	2.6	2.7	2.9	3.2
learned_b	4.2	4.5	4.9	5.4
learned_a,b	4.1	4.5	4.9	5.4
MLP	6.4	6.3	7.0	7.2

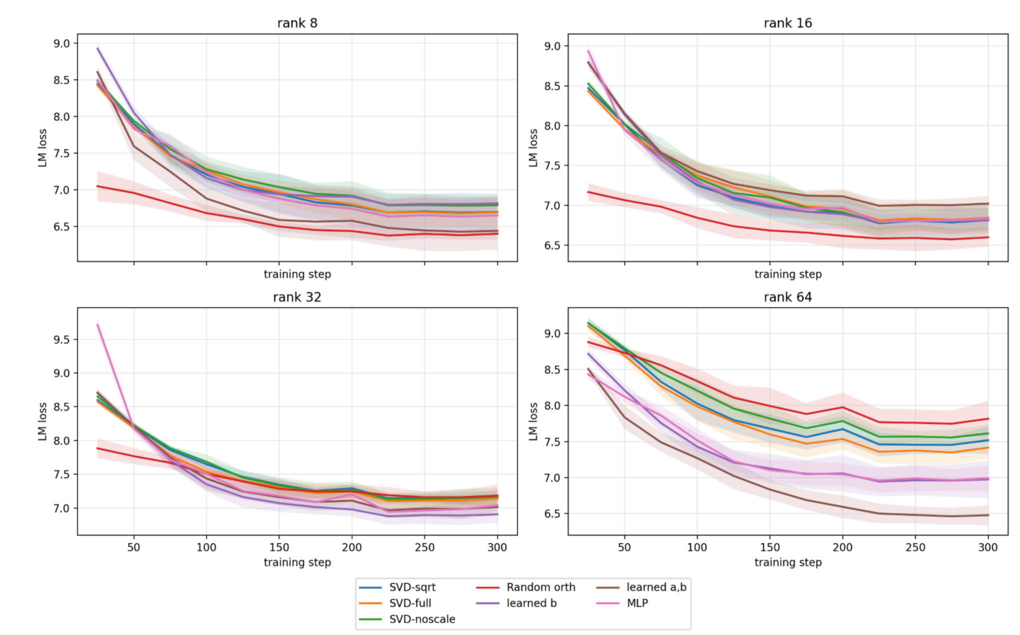
Approx. log condition numbers; lower is better.

Results

- Learned strategies, especially MLP, achieve lower proxy MSE but worse conditioning.
- Random orthogonal has the best conditioning but highest proxy MSE.
- This suggests a tradeoff between predictive quality and numerical stability.

Experiments/Results - Next-Token Prediction

- Use the proxy as a cheap model-selection stage for learned strategies.
- Reuse learned scaling parameters in next-token training.
- **Goal**: Replace Pythia Layer 2 attention with each SKA initialization, freeze other layers, and evaluate LM loss/perplexity.



Rank	Best method	Loss/PPL
8	Random orth.	6.09 / 446.6
16	Random orth.	6.21 / 513.6
32	learned_b	6.78 / 887.0
64	learned_a,b	6.47 / 655.6

Best next-token strategy by rank. Lower is better.

Figure 2. Next-token training curves for each strategy.

Results

- Random orthogonal is best at low ranks $(8, 16)$.
- Learned methods outperform fixed SVD baselines at higher ranks.
- Low ranks favor conditioning; higher ranks preserve more W_K geometry.

Conclusion/Future Work

Primary Findings:

- Winning high-quality Koopman prediction quality alone does not make better initialization: conditioning matters as well.
- Learned initialization isn't universally better; it helps when there's enough capacity to use the structure of the original attention weights.

Future Work:

- Test larger models, replace multiple layers, study whether ideal strategy changes with depth
- Attempt to train learned scalings directly on next-token LM-loss rather than through a cheaper proxy

References

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[3] Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. Pointer sentinel mixture models. In *ICLR*, 2017.

[4] Anupama Sridhar and Alexander Johansen. Echo: Kv-cache-free associative recall with spectral koopman operators, 2026.